

DailyTest

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```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyr)
library(UsingR)
```

```
## Loading required package: MASS

##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##   select

## Loading required package: HistData

## Loading required package: Hmisc

##
## Attaching package: 'Hmisc'

## The following objects are masked from 'package:dplyr':
##
##   src, summarize

## The following objects are masked from 'package:base':
##
##   format.pval, units
```

1. What categories are there?

There are ways to check how many unique values there are in the Groups section of the data. None of them are as slow as trying to re-learn R from first principles, but none of them get me to re-learn R from first principles.

To start, the table:

```
jons_guest_list <- read.csv("C:/Users/Grad School/Desktop/RCode/Data/daily_show_guests.csv",
                           header = TRUE)
head(jons_guest_list, 10)
```

##	YEAR	GoogleKnowlege_Occupation	Show	Group	Raw_Guest_List
## 1	1999	actor	1/11/99	Acting	Michael J. Fox
## 2	1999	Comedian	1/12/99	Comedy	Sandra Bernhard
## 3	1999	television actress	1/13/99	Acting	Tracey Ullman
## 4	1999	film actress	1/14/99	Acting	Gillian Anderson
## 5	1999	actor	1/18/99	Acting	David Alan Grier
## 6	1999	actor	1/19/99	Acting	William Baldwin
## 7	1999	Singer-lyricist	1/20/99	Musician	Michael Stipe
## 8	1999	model	1/21/99	Media	Carmen Electra
## 9	1999	actor	1/25/99	Acting	Matthew Lillard
## 10	1999	stand-up comedian	1/26/99	Comedy	David Cross

With that code having run, we can see the types of data in the file:

```
colnames(jons_guest_list)
```

```
## [1] "YEAR"                "GoogleKnowlege_Occupation"
## [3] "Show"                "Group"
## [5] "Raw_Guest_List"
```

From there, I'm interested in the Groups:

```
jons_groups <- jons_guest_list$Group
jons_groups <- as.data.frame(jons_groups)
jons_groups <- distinct(jons_groups)
jons_groups
```

##	jons_groups
## 1	Acting
## 2	Comedy
## 3	Musician
## 4	Media
## 5	<NA>
## 6	Politician
## 7	Athletics
## 8	Business
## 9	Advocacy
## 10	Political Aide
## 11	Misc
## 12	Academic

```
## 13      Government
## 14      media
## 15      Clergy
## 16      Science
## 17      Consultant
## 18      Military
```

Now we know what the different groups are. Yay!

2. Plot change over time

I don't care about any data beyond the year and group:

```
jons_guest_list <- subset(jons_guest_list, select = c("YEAR", "Group"))
head(jons_guest_list, 10)
```

```
##   YEAR   Group
## 1 1999   Acting
## 2 1999   Comedy
## 3 1999   Acting
## 4 1999   Acting
## 5 1999   Acting
## 6 1999   Acting
## 7 1999 Musician
## 8 1999   Media
## 9 1999   Acting
## 10 1999  Comedy
```

Rotating that so we have a new in which each column is a year, each row is a profession (Group), and that intersection is how many of that profession Jon had on that year:

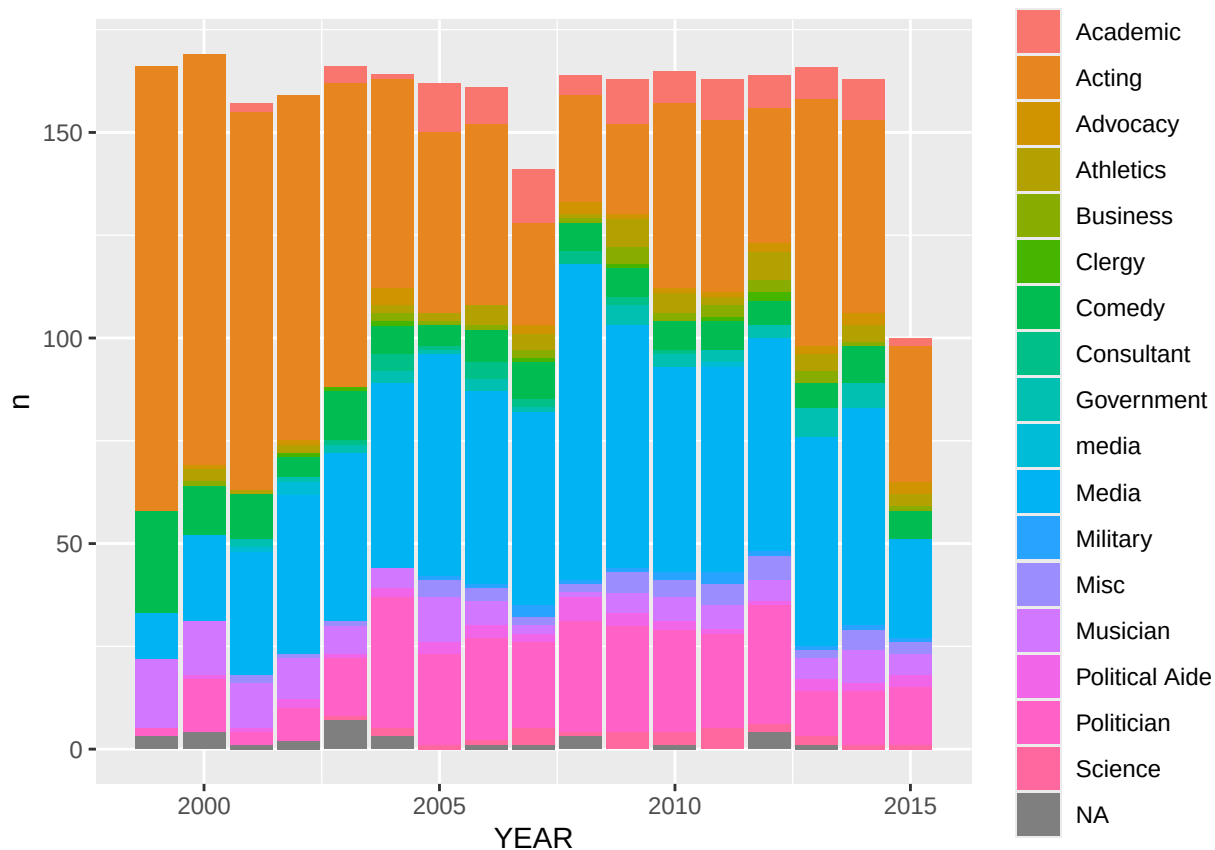
```
jons_guest_list <- count(jons_guest_list, Group, YEAR)
#jons_guest_list
# jons_guest_list <- pivot_wider(jons_guest_list, names_from = YEAR,
#                               #values_from = n, values_fill = 0)

head(jons_guest_list, 10)
```

```
##      Group YEAR  n
## 1 Academic 2001  2
## 2 Academic 2003  4
## 3 Academic 2004  1
## 4 Academic 2005 12
## 5 Academic 2006  9
## 6 Academic 2007 13
## 7 Academic 2008  5
## 8 Academic 2009 11
## 9 Academic 2010  8
## 10 Academic 2011 10
```

So with all that done, I should be able to finally build this graph:

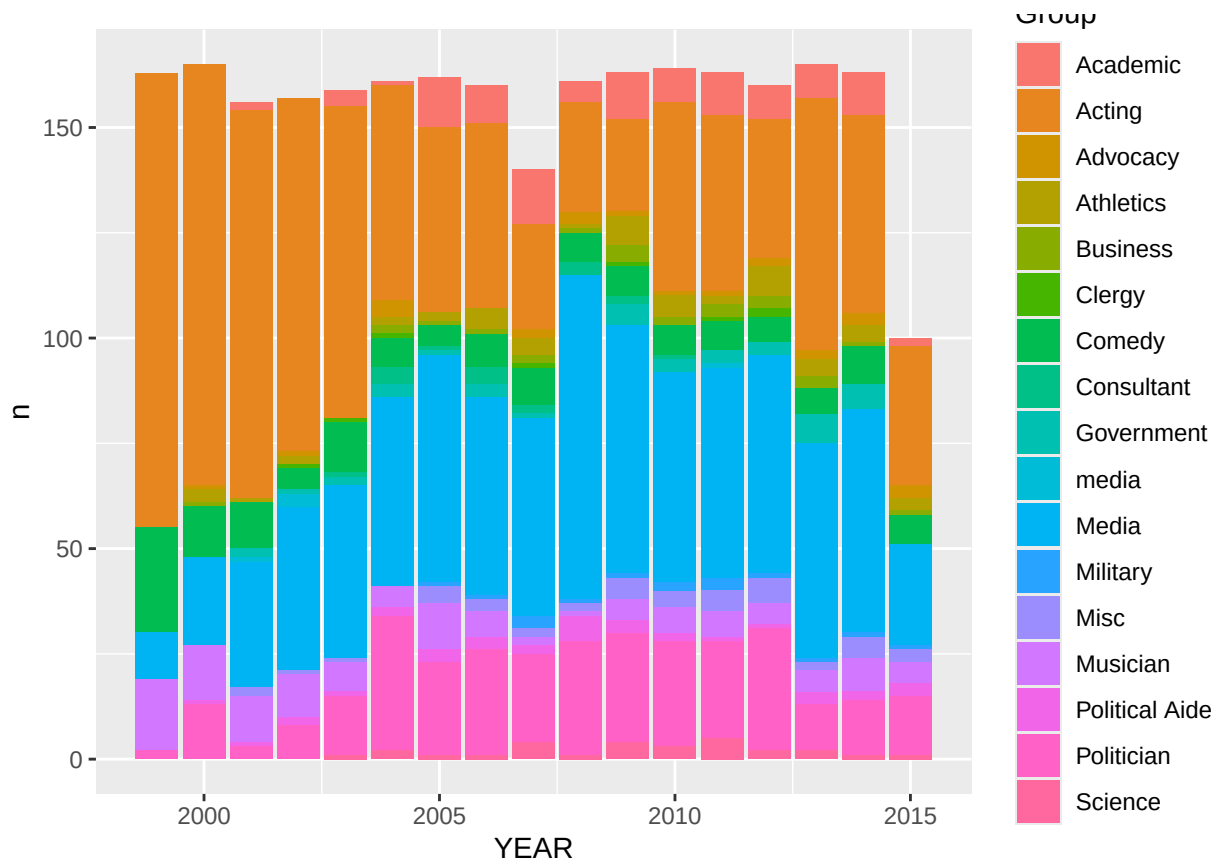
```
ggplot(jons_guest_list, aes(fill=Group, y=n, x=YEAR)) +  
  geom_bar(position="stack", stat="identity")
```



I skipped a data cleaning step earlier, so I'm making up for it now. Namely: 1. "N/A" indicates no guest should be discounted; 2. "media" and "Media" should be the same under "Media"; 3. "Academic" and "Science" can be merged; 4. "Advocacy", "Business", "Clergy", "Consultant", "Military", and "Political Aide" can all go under Misc. 5. The chart should be normalized because if I'm comparing how popular politicians v comedians are, it isn't helpful to say they both went down in popularity by about half when Jon left the show halfway through the year. 6. The y-axis should be called something sensible.

For the N/A:

```
jons_guest_list <- filter(jons_guest_list, Group != "N/A")  
ggplot(jons_guest_list, aes(fill=Group, y=n, x=YEAR)) +  
  geom_bar(position="stack", stat="identity")
```

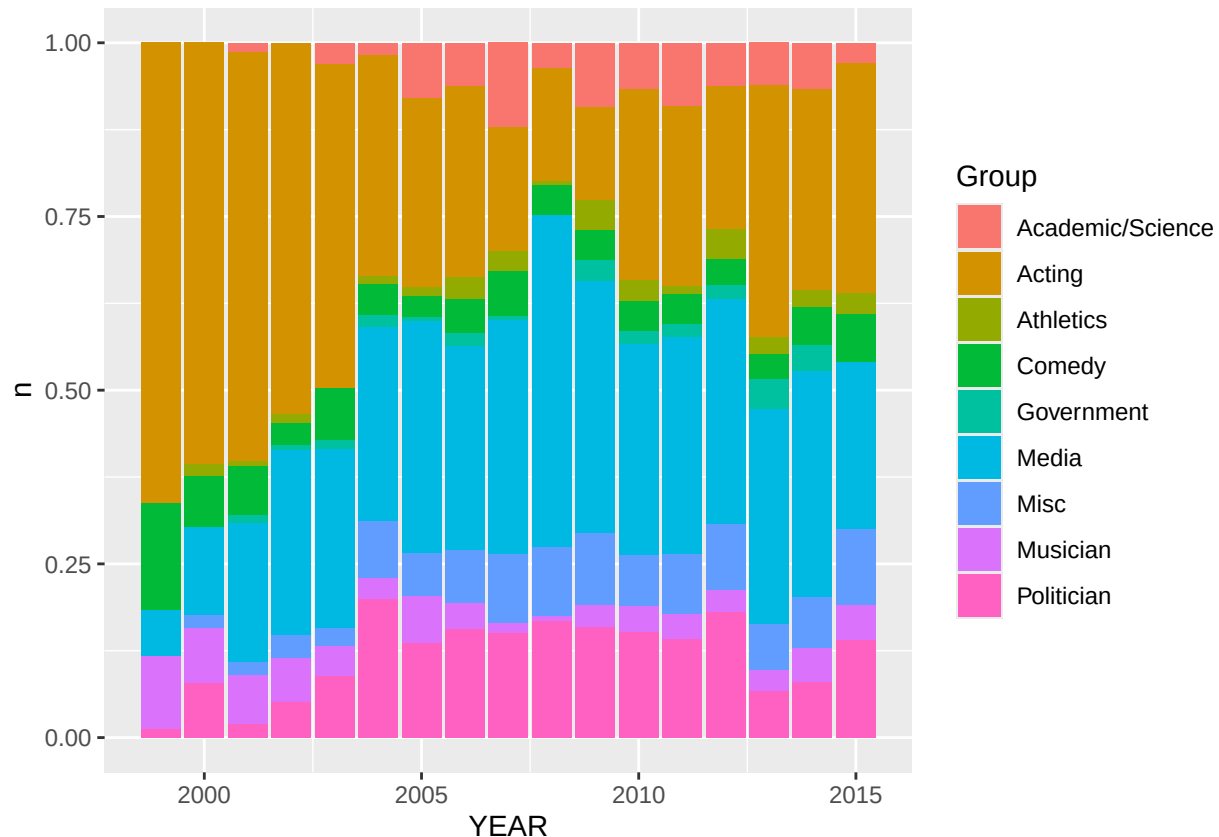


For the rest, it's easier to just start with clean data

```
jons_guest_list_2 <- read.csv("C:/Users/Grad School/Desktop/RCode/Data/daily_show_guests.csv",
                             header = TRUE)
jons_guest_list_2 <- filter(jons_guest_list_2, Group != "N/A")
jons_guest_list_2$Group[jons_guest_list_2$Group == "media"] <-
  "Media"
jons_guest_list_2$Group[jons_guest_list_2$Group == "Academic"] <-
  "Academic/Science"
jons_guest_list_2$Group[jons_guest_list_2$Group == "Science"] <-
  "Academic/Science"
jons_guest_list_2$Group[jons_guest_list_2$Group == "Advocacy"] <-
  "Misc"
jons_guest_list_2$Group[jons_guest_list_2$Group == "Business"] <-
  "Misc"
jons_guest_list_2$Group[jons_guest_list_2$Group == "Clergy"] <-
  "Misc"
jons_guest_list_2$Group[jons_guest_list_2$Group == "Consultant"] <-
  "Misc"
jons_guest_list_2$Group[jons_guest_list_2$Group == "Military"] <-
  "Misc"
jons_guest_list_2$Group[jons_guest_list_2$Group == "Political Aide"] <-
  "Misc"
jons_guest_list_2 <- subset(jons_guest_list_2, select = c("YEAR", "Group"))
jons_guest_list_2 <- count(jons_guest_list_2, Group, YEAR)

ggplot(jons_guest_list_2, aes(fill=Group, y=n, x=YEAR)) +
```

```
geom_bar(position="fill", stat="identity")
```



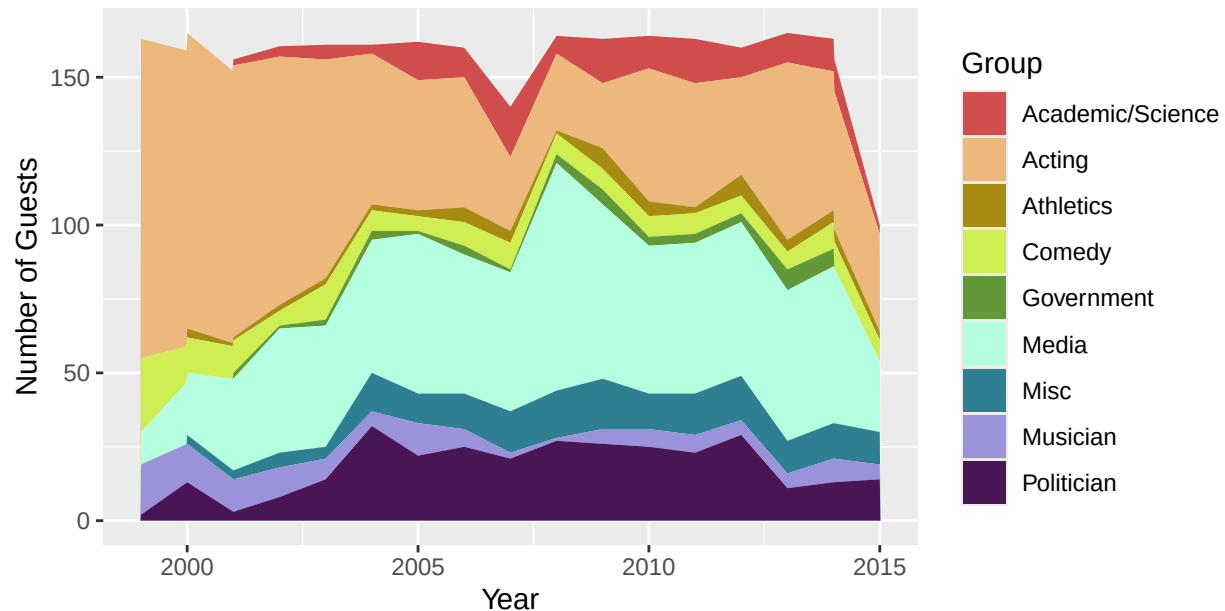
So that's basically what I want. To make this a stacked line chart, and then add some custom coloring, and then actually deciding to not normalize the data:

```
custom_colors = c("#d14d4b", "#ecb87c", "#a78c11", "#cfee51", "#629938", "#b5ffe1", "#2d7f91", "#9a92da")

ggplot(jons_guest_list_2, aes(fill=Group, y=n, x=YEAR)) +
  geom_area() +
  labs(
    title = "The Types of Guests on Jon Stwert's Daily Show Over Time",
    subtitle = "Data procured from FiveThirtyEight",
    caption = "Graph by Rainier Pederson. Data from FiveThirtyEight,\n
    found at https://github.com/fivethirtyeight/data/tree/master/daily-show-guests.\n
    Used under CC by 4.0.",
    tag = waiver(),
    dictionary = waiver(),
    alt = waiver(),
    alt_insight = waiver(),
    x = "Year",
    y = "Number of Guests"
  ) +
  scale_fill_manual(values = custom_colors)
```

The Types of Guests on Jon Stwert's Daily Show Over Time

Data procured from FiveThirtyEight



Graph by Rainier Pederson. Data from FiveThirtyEight,

found at <https://github.com/fivethirtyeight/data/tree/master/daily-show-guests>.

Used under CC by 4.0.

There are weird jumps that correspond to a lack of data where data should have been set to 0. This wasn't caught in the bar graph stage because there it wouldn't have mattered. To clean that:

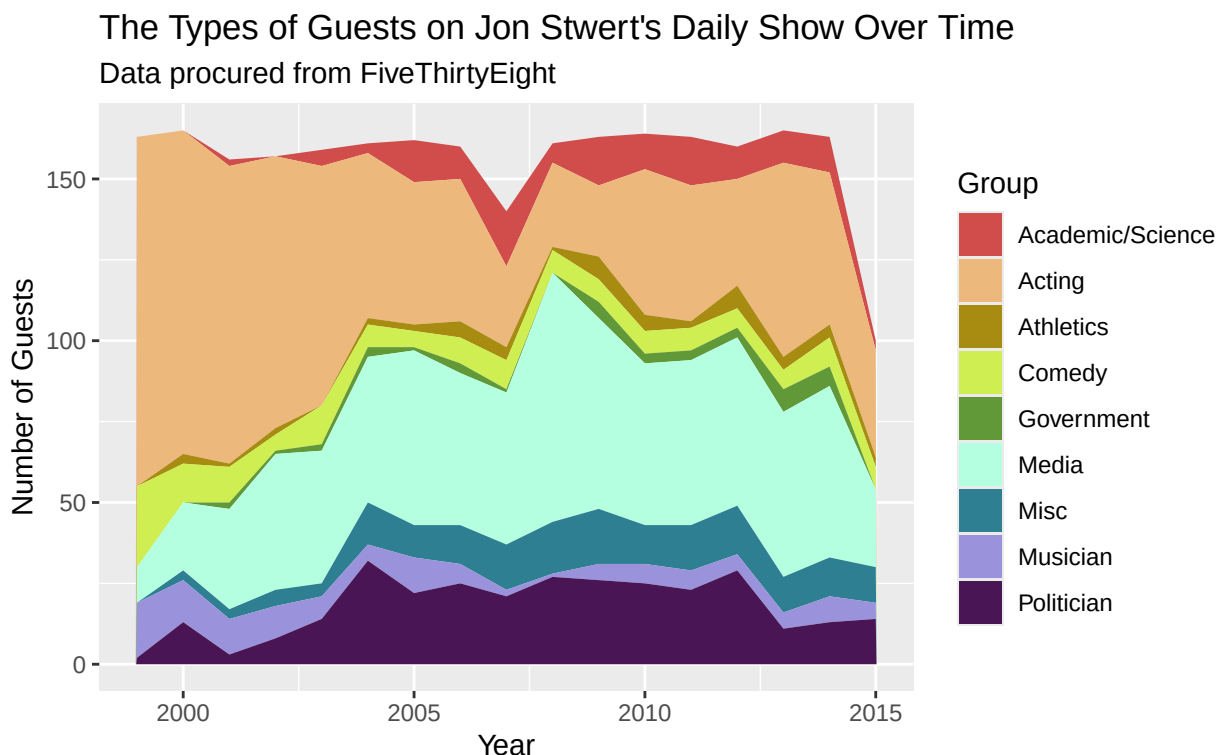
```
head(jons_guest_list_2, 10)
```

```
##           Group YEAR  n
## 1 Academic/Science 2001  2
## 2 Academic/Science 2003  5
## 3 Academic/Science 2004  3
## 4 Academic/Science 2005 13
## 5 Academic/Science 2006 10
## 6 Academic/Science 2007 17
## 7 Academic/Science 2008  6
## 8 Academic/Science 2009 15
## 9 Academic/Science 2010 11
## 10 Academic/Science 2011 15
```

```
jons_guest_list_3 <- jons_guest_list_2
jons_guest_list_2 <- pivot_wider(jons_guest_list_2, names_from = YEAR,
                                values_from = n, values_fill = 0)
jons_guest_list_2 <- pivot_longer(jons_guest_list_2, !Group, names_to = "YEAR")
# print(as_tibble(jons_guest_list_2), n = 200)
# Yes, the 0s are

# jons_guest_list_2 <- as.data.frame(jons_guest_list_2)
jons_guest_list_2$YEAR <- as.integer(jons_guest_list_2$YEAR)
```

```
ggplot(jons_guest_list_2, aes(fill=Group, y=value, x=YEAR)) +
  geom_area() +
  labs(
    title = "The Types of Guests on Jon Stwert's Daily Show Over Time",
    subtitle = "Data procured from FiveThirtyEight",
    caption = "Graph by Rainier Pederson. Data from FiveThirtyEight,\n found at https://github.com",
    tag = waiver(),
    dictionary = waiver(),
    alt = waiver(),
    alt_insight = waiver(),
    x = "Year",
    y = "Number of Guests"
  ) +
  scale_fill_manual(values = custom_colors)
```



Graph by Rainier Pederson. Data from FiveThirtyEight,
found at <https://github.com/fivethirtyeight/data/tree/master/daily-show-guests>.
Used under CC by 4.0.

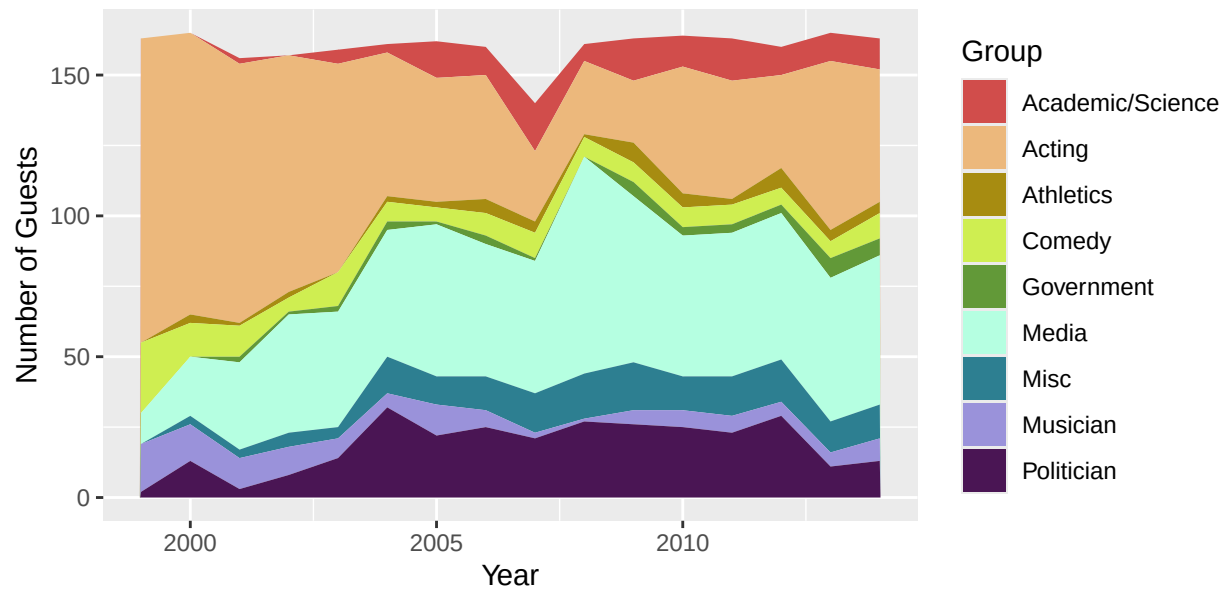
That took a minute (week) to work. As it turns out, when `jons_guest_list_2` was pivoting and un-pivoting, the `YEAR` went from int to char and that caused a whole bunch of problems.

Anyways, my last thing is to run the data but without including 2015 because visually, the sudden decrease is throwing me.

The Types of Guests on Jon Stwert's Daily Show Over Time

From 1999–2014

Data procured from FiveThirtyEight



Graph by Rainier Pederson. Data from FiveThirtyEight,
found at <https://github.com/fivethirtyeight/data/tree/master/daily-show-guests>.
Used under CC by 4.0.

And finally, exporting just the image to show my dad:

```
ggsave("DailyGuests1999-2014.png")
```

```
## Saving 6.5 x 4.5 in image
```

```
## Warning in grDevices::dev.off(): agg could not write to the given file
```

And with all that, I think this has been a successful first project in getting back into R! :)